Best Practices for Biomedical Research Data Management

Module 2: Research Lifecycle Research Workflow

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This video will give an overview of the data lifecycle in more detail from the perspective of a Chemistry department Postdoctoral Researcher and highlight their daily workflow in the laboratory and field setting.

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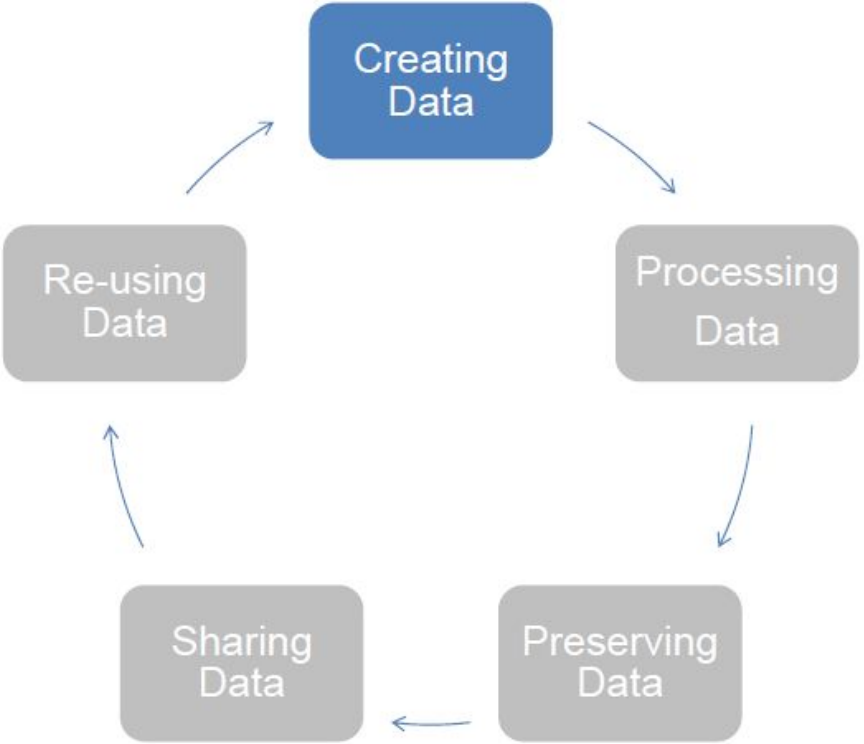
Workflow: The Research & Data Lifecycle



1. Design Experiment
   - Methods
   - Sample Login
2. Collect Raw Data
   - Experiment
   - Observations
   - Measurements
3. Capture metadata

Adapted from http://www.data-archive.ac.uk/create-manage/life-cycle

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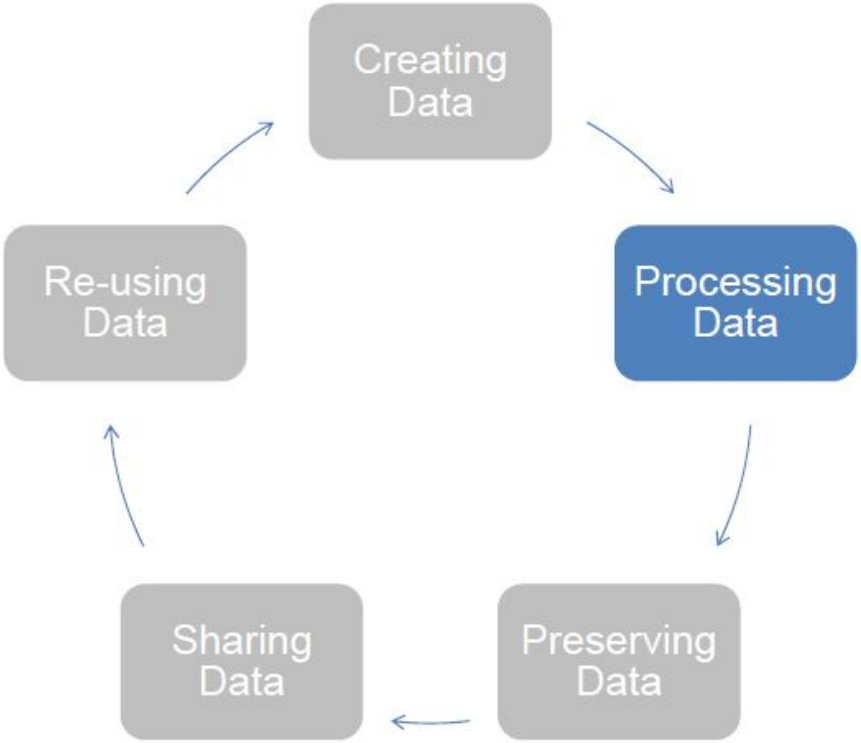

1. Interpret Data
   - Tables
   - Charts
   - Graphs
   - Drawings
   - Pictures
2. Review Data
   - PI
   - Collaborators
   - Regulators
3. Prepare Manuscript

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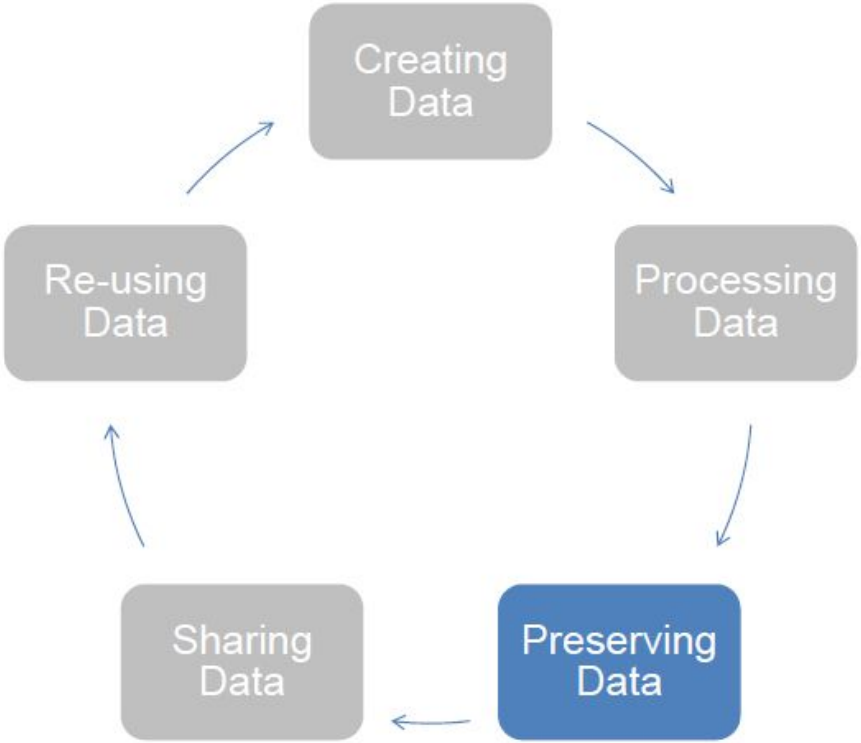

1. Migrate Data
   - Format
2. Backup/Store Data
   - Local
   - Network
   - Repository
3. Create Metadata, Documentation
4. File/ Folder Organizational Scheme
   - Concise, meaningful
   - Consistent
   - Versioning

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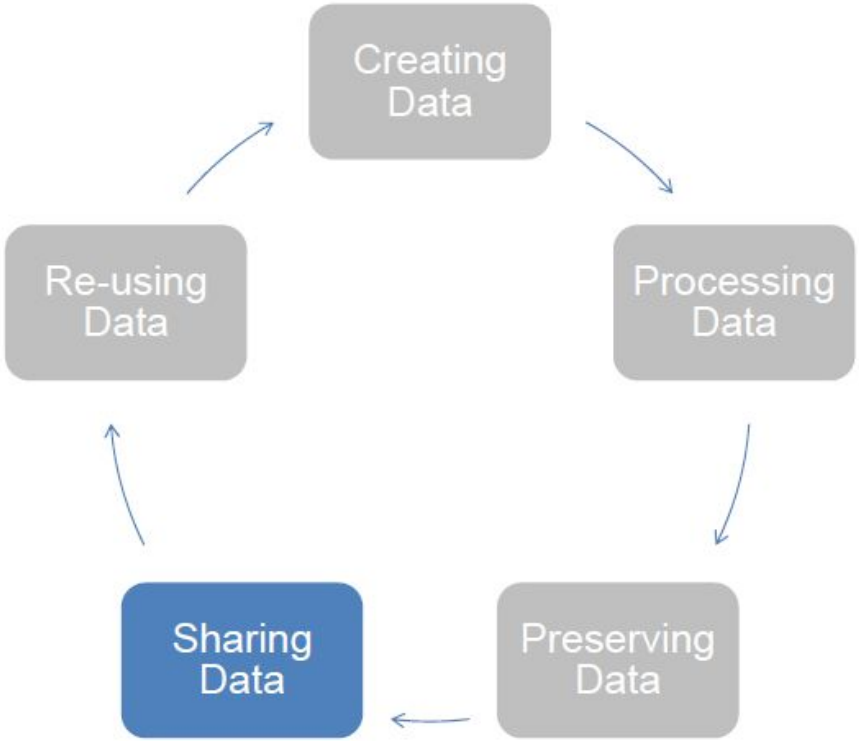

1. Share w/ Collaborators
2. Promote Data
3. Control Access
   - Secure method
   - DOI

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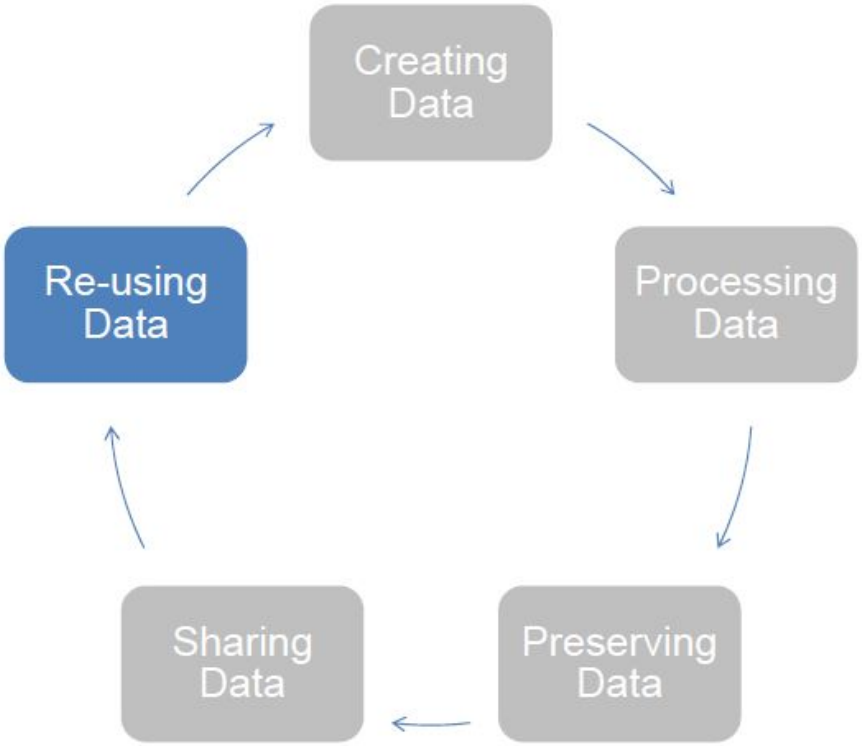

1. New Research
2. Teach/Learn
3. Follow-up Research

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